

# NC(SMC)<sup>2</sup> 2026 Speed Round

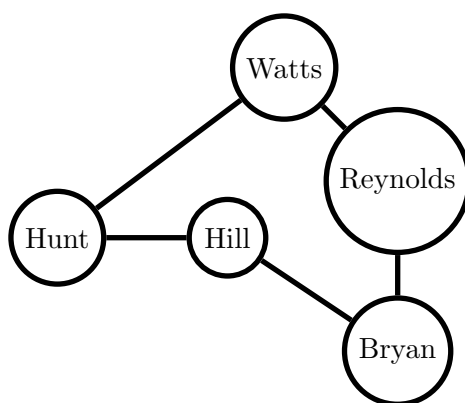
NC(SMC)<sup>2</sup> Problem Writers

May 9th, 2026

1. Let  $a \# b = ab^2 + ba^2$  and  $a \& b = ab^2 - ba^2$ . What is  $(3 \# 2) \& (2 \# 3)$ ?
2. One day, Nelson suddenly has a boba craving. He walks 8 miles North and 3 miles East. He realizes that he is in the wrong place and walks another 4 miles South to the boba shop. How far is he from his original position?
3. Michael needs to freshen up for his first date. To do this, he must complete his morning routine of drinking 2 glasses of orange juice every 3 times he brushes his teeth. If Michael brushed his teeth 27 times that day, how many glasses of orange juice did he drink?
4. The Chicken Alex is a youtuber with 2 chickens. He's doing a giveaway to see which of his 39 subscribers will receive his chickens. The Chicken Alex gives them away one at a time independently to a random subscriber. The probability that one of The Chicken Alex's subscribers will receive both chickens can be written as  $\frac{p}{q}$  in simplest form. What is  $p + q$ ?
5. What is the average measure of the exterior angles of a decagon (10-sided polygon) in degrees?
6. Yae likes to dress up her two identical Mei dolls. She has 4 different outfits and 3 different hats. She gives each doll an outfit and a hat. How many different dress up combinations can she make?
7. What integer number other than 0 satisfies the property that five times the number minus its square equals itself?
8. How many ways are there to arrange the numbers 1, 2, 3, 4, 5 in a line such that no number is next to a number that is more than double itself?
9. Jack, Robert, and Wyatt are playing a game of frisbee. If Jack throws it to Wyatt for a gain of twelve yards and then Wyatt throws it to Robert for a fifty-four yard gain, what percent of the total length field did they cover? A frisbee field has two twenty yard endzones separated by 70 yards.

10. For Pi Day, Susan bought a pie to share with her friends Abby, Angela, and Jessica. Susan goes first and takes one third of the pie. Abby goes second and takes two fifths of what's left. Then Angela takes two thirds of what's left and gives the rest to Jessica. If the fraction of the pie Jessica gets can be expressed as  $\frac{m}{n}$  in simplest form, find  $m + n$ .
11. Dr. Leaving enters the positive integer  $x$  into a calculator. Then he presses  $+x$  repeatedly for a total of  $x$  times. Next, he presses  $\div x$ . Finally, he presses  $=$  and the result is 2026. What is  $x$ ?
12. Yo lives on a perfectly flat Earth. Yo, who lives in Atlantis, travels 300 miles to Forbodulon Prime, then 390 miles to Zootopia 2, then 390 miles back to Atlantis. All of his flights are perfectly straight. If every point of land and sea strictly in the triangle formed by Yo's flights falls into the endless void of space, what is the total surface area of Yo's Earth that was lost?
13. What is the fourth smallest even positive integer with an odd number of divisors?
14. Alex, Ben, and Carson are in a tournament in which each pair of players plays one game. Alex has a  $\frac{1}{3}$  chance of winning against Ben. Ben has a  $\frac{1}{4}$  chance of winning against Carson. Carson has a  $\frac{1}{2}$  chance of winning against Alex. The probability that each player ends up with exactly one win can be represented as  $\frac{m}{n}$  in simplest form. What is  $m + n$ ?
15. A unicorn and a dragon start at opposite ends of a 150 meter long track. They race toward each other. The dragon flies at a speed of 7 meters per second while the unicorn gallops at 8 meters per second. How many seconds does it take for them to meet?
16. The mean, median, and unique mode of a set of 6 positive integers is 7. What is the largest number that can be in the set?
17. Carter throws a frisbee 20 meters to Alan. Alan tries to throw the frisbee back, but it ends up going 20 meters in a random direction. The probability that the frisbee ends up less than 20 meters away from Carter can be written in simplest form as  $\frac{m}{n}$ . What is  $m + n$ ?
18. What is the area of region in the  $xy$ -plane enclosed by  $y = |x + 2| - 2$  and  $y = -|x + 2| + 2$ ?
19. The smallest sides of three isosceles right triangles form an arithmetic sequence. If the smallest triangle has area 8, and the smallest side of the largest triangle is 8, what is the sum of the areas of all three triangles?
20. If 3 numbers  $A$ ,  $B$ , and  $C$  have pairwise sums of 41, 83, and 110, what is the sum of the digits of  $A + B + C$ ?

21. If real numbers  $x$  and  $y$  satisfy  $x + 1013y = 2026$ , what is the maximum value of  $xy$ ?
22. Square  $ABCD$  shares one side with the equilateral triangle  $AEB$  ( $E$  is in the exterior of  $ABCD$ ). What is the degree measure of angle  $CED$ ?
23. What is the smallest number made up of only 6's and 7's that is divisible by 88?
24. Equilateral triangles are erected on each side of a square of side length 6, forming the net of a square pyramid. If the sides of the square are folded together, the volume of the pyramid can be expressed as  $a\sqrt{b}$  in simplest form. Compute  $a + b$ .
25. There is a sequence of 10 elements  $(a_1, a_2, \dots, a_{10})$ . Srinivas sets each  $a_i$  to either 1 or  $-1$  at random. He then sums up the elements from  $a_1$  to  $a_{10}$  in order. For example, in this sequence of 5 elements  $(1, -1, -1, -1, 1)$ , the sums after each element in the process are  $1, 0, -1, -2, -1$ . The probability that the current sum is either  $-1, 0$ , or  $1$  at all points in this process can be expressed as  $\frac{m}{n}$  in simplest form. What is  $m + n$ ?
26. Right triangle  $ABC$  has a right angle at  $\angle C$ . Let  $M$  and  $N$  be the midpoints of  $\overline{BC}$  and  $\overline{CA}$  respectively. The medians  $\overline{AM}$  and  $\overline{BN}$  lengths of 10 and  $4\sqrt{10}$ . What is the area of triangle  $ABC$ ?
27. A math student and physics student take turns incrementing a number, which starts at 0. In the first minute, the math student adds 8 to the number. In the second minute, the physics student adds 9. They continue taking turns adding 8 and 9, alternating every minute, until the number is divisible by 50. How many minutes will they increment the number before stopping?
28. Steven is going from his dorm in Hunt to his classes in Bryan. He can go directly from Hunt to Watts, from Watts to Reynolds, from Reynolds to Bryan, from Bryan to Hill, from Hill to Hunt, and vice versa, in 1 minute each. How many ways are there for Steven to go from Hunt to Bryan in exactly 9 minutes, assuming he moves every minute? Note: you can go through Bryan and other places multiple times, but must end up in Bryan.



29. Four spheres with radii 1, 2, 3, 4 are arranged such that each sphere is externally tangent to at least one of the other spheres. Additionally, there exists a path along the surfaces of the spheres connecting any two of the spheres. What is the maximum possible volume of the tetrahedron formed by the centers of each of the spheres?
30. Let  $Q, R, S$  be integers such that  $Q + R + S = -24$  and  $QRS - Q^2 - R^2 - S^2 = -560$ . What is the largest value among  $Q, R$ , and  $S$ ?